

Heisenberg 解析への言秀(1)

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(宣伝)

- CR多様体の埋め込み可能性
- CR多様体に関する個人的諸問題

(参考文献)

- Folland - Stein 1974,
Estimates for the $\bar{\partial}_b$ -complex and
analysis on the Heisenberg group
- Fischer - Ruzhansky 2016,
Quantization on nilpotent Lie groups
- Ponge 2008,
Heisenberg calculus and spectral
theory of hypoelliptic operators on
Heisenberg manifolds

§0 Introduction

代表的な linear PDO

(Riemannian / conformal geometry)

$$\Delta, L = \Delta + \frac{n-1}{4(n-2)} R$$

Pancitz op, GJMS op

) elliptic

(CR geometry)

$$\Delta_b, \square_b, L = \Delta_b + \frac{n}{2(n+1)} R,$$

CR Pancitz op, CR GJMS op

) not elliptic

→ 解決策: Heisenberg calculus

(theory of HΨDO)

§1 Analysis on the Heisenberg group

Heisenberg group : $\mathbb{H}^n = \mathbb{C}^n \times \mathbb{R}$
 $\downarrow$
 (z, t)

group str : $(z, t) \cdot (z', t')$
 $= (z + z', t + t' + 2\operatorname{Im}(z \cdot \bar{z}'))$

dilation : $S_r(z, t) = (rz, r^2 t) \quad (r > 0)$
: group hom

$\theta := dt + i \sum_{j=1}^n (z_j d\bar{z}_j - \bar{z}_j dz_j)$
: left-inv contact form

$T = \frac{\partial}{\partial t}$: Reeb v.f. (bi-inv)

$Z_j := \frac{\partial}{\partial z_j} + i \bar{z}_j \frac{\partial}{\partial t}$: left-inv v.f.

$T^{1,0} \mathbb{H}^n := \operatorname{Span} \{Z_1, \dots, Z_n\}$

: strictly pseudoconvex CR str on $\mathbb{H}^n$

$$\mathcal{L}_\alpha := -\frac{1}{2} \sum_{j=1}^n (z_j \bar{z}_j + \bar{z}_j z_j) + i \alpha T$$

: Folland-Stein op ($\alpha \in \mathbb{R}$)

この op に関する可解性は

基本解 ($\mathcal{L}_\alpha \Phi_\alpha = \delta$) と結びついていゝ。

$$p(z, t) := |z|^2 - it : \mathbb{C}R \text{ hol, hom of deg 2}$$

$$\varphi_\alpha = p^{-\frac{1}{2}(n+\alpha)} \bar{p}^{-\frac{1}{2}(n-\alpha)}$$

$$\in L^1_{loc} \cap C^\infty(\mathbb{H}^n \setminus \{(0,0)\})$$

: hom of deg $-2n$

$$\text{この } \varphi_\alpha \text{ に対し } \mathcal{L}_\alpha \varphi_\alpha = C_\alpha \delta, \text{ かつ}$$

$$C_\alpha = \frac{2^{2-n} \pi^{n+1}}{\Gamma\left(\frac{n+\alpha}{2}\right) \Gamma\left(\frac{n-\alpha}{2}\right)} = \int_{\mathbb{H}^n} \mathcal{L}_\alpha \varphi_\alpha d\mu$$

$$\left(\begin{array}{l} p \text{ を } p + \varepsilon^2 \text{ としてものを } \varphi_{\alpha, \varepsilon} \text{ とする,} \\ \mathcal{L}_\alpha \varphi_{\alpha, \varepsilon} \rightarrow \mathcal{L}_\alpha \varphi_\alpha, \text{ supp } \mathcal{L}_\alpha \varphi_\alpha = \{(0,0)\}, \\ C_\alpha = \int_{\mathbb{H}^n} \mathcal{L}_\alpha \varphi_{\alpha, \varepsilon} d\mu : \text{ indep of } \varepsilon. \end{array} \right)$$

特に $C_\alpha = 0$ となるのは $\alpha \in \pm(n+2\mathbb{Z}_{\geq 0})$.

$\alpha \notin \pm(n+2\mathbb{Z}_{\geq 0})$ のとき,

$\Phi_\alpha := C_\alpha^{-1} \varphi_\alpha$ は $\mathcal{L}_\alpha \Phi_\alpha = \delta_0$ をみたす.

これを用いると, $f \in C_c^\infty$ に対して

$$\mathcal{L}_\alpha(f * \Phi_\alpha) = f * \delta_0 = f \text{ が分かる.}$$

[Thm (FS, Proposition 7.5)

$$\mathcal{L}_\alpha : \text{hypoelliptic} \iff \alpha \notin \pm(n+2\mathbb{Z}_{\geq 0})$$

⊙ $\alpha \in \pm(n+2\mathbb{Z}_{\geq 0})$ のとき,

$\varphi_\alpha \notin C^\infty$ だが $\mathcal{L}_\alpha \varphi_\alpha = 0$ である.

$\alpha \notin \pm(n+2\mathbb{Z}_{\geq 0})$ のとき,

$\Phi_\alpha \in C^\infty(\mathbb{H}^n \setminus \{(0,0)\})$ であることが

$\mathcal{L}_\alpha : \text{hypoelliptic}$ がしたがう.

[c.f. Folland, Intro to PDE, Thm 1.58)

□

§2 Relation to representation theory

$\lambda \in \mathbb{R}^x$ に対して, *Schrödinger 表現*

$\pi_\lambda: \mathcal{H}^n \rightarrow \mathcal{U}(L^2(\mathbb{R}^n))$ が

$$(\pi_\lambda(x + iy, t) u)(\xi)$$

$$= e^{i\lambda(t + 2x \cdot y + 2y \cdot \xi)} u(\xi + 2y)$$

で定まる. このとき

$$d\pi_\lambda(\mathcal{L}_\alpha) = \underline{\Delta + \lambda^2 |\xi|^2} - \alpha \lambda$$

Hamiltonian of QHD

$$\text{Spec } d\pi_\lambda(\mathcal{L}_\alpha) = \{ |\lambda| (n + 2k) - \alpha \lambda \mid k \in \mathbb{Z}_{\geq 0} \}$$

特に

$$\alpha \notin \pm(n + 2\mathbb{Z}_{\geq 0})$$

$$\iff \forall \lambda \in \mathbb{R}^x, \forall u \in L^2(\mathbb{R}^n),$$

$$d\pi_\lambda(\mathcal{L}_\alpha) u = 0 \Rightarrow u = 0$$

(*Rockland 条件*)

(Group Fourier transform)

$f \in \mathcal{S}$ に対して,

$$\hat{f}(\lambda) := \int_{\mathbb{H}^n} f(z, t) \pi_\lambda(z, t)^* d\mu$$

$$\in B_2(L^2(\mathbb{R}^n)) \leftarrow \text{sp of HS ops}$$

の性質

$$\cdot \widehat{f_1 * f_2}(\lambda) = \hat{f}_2(\lambda) \hat{f}_1(\lambda)$$

$$\cdot \widehat{\mathcal{L}_\alpha f}(\lambda) = d\pi_\lambda(\mathcal{L}_\alpha) \hat{f}(\lambda)$$

$$\cdot f(z, t) = \frac{2^{n-1}}{\pi^{n+1}} \int_{\mathbb{R}^n} \text{tr}(\pi_\lambda(z, t) \hat{f}(\lambda)) |\lambda|^n d\lambda$$

(inversion formula)

これを踏まえて、基本解 Φ_α は

$$\Phi_\alpha(z, t)$$

$$= \frac{2^{n-1}}{\pi^{n+1}} \int_{\mathbb{R}^n} \text{tr}(\pi_\lambda(z, t) d\pi_\lambda(\mathcal{L}_\alpha)^{-1}) |\lambda|^n d\lambda$$

であることが分かる。

§3 Introduction to Heisenberg calculus

P : linear PDO on $(M^{2n+1}, T^{1,0}M)$

$\forall a \in M, \exists P_a$: hom left-inv linear PDO
"principal part of P at a "

$\leadsto d\pi_\lambda(P_a)$: "principal symbol"

Thm (P , Theorem 3.3.18)

$\exists Q$: HPDO s.t. $PQ \sim QP \sim I$

$\Leftrightarrow \forall a \in M, P_a \subset P_a^*$ は Rockland 条件をみたす

$\Leftrightarrow \forall a \in M, \forall \lambda \in \mathbb{R}^\times, \forall u \in L^2(\mathbb{R}^n),$

$$d\pi_\lambda(P_a)u = 0$$

or

$$d\pi_\lambda(P_a^*)u = 0$$

$$\Rightarrow u = 0$$

(Application)

$\alpha \notin \pm(n+2\mathbb{Z}_{>0})$ のとき, $\Delta_b + i\alpha T$ の

スペクトルの挙動は elliptic のときと同じ.

($\odot$ principal part は L_α で"ある".)